

Implantable MIMO Antenna for WMTS and ISM Bands in Leadless Cardiac Pacemaker System

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Abstract—This study presents a compact dual-band implantable multi-input multi-output (MIMO) antenna for leadless cardiac pacemaker applications. The proposed antenna operates within the Wireless Medical Telemetry Service (WMTS) band (1.395–1.4 GHz) and the Industrial, Scientific, and Medical (ISM) band (2.4–2.4835 GHz). To achieve miniaturization, a meandered structural configuration is employed, resulting in an overall compact dimension of $8 \times 6.4 \times 0.5$ mm³. The antenna is encapsulated in implantable biomedical device (IMD) and evaluated in both homogeneous and heterogeneous heart phantom simulation environments. Performance indicate optimal characteristics, including reflection coefficients of -29.29 dB and -29.48 dB, gains of -26.3 dBi and -30.15 dBi, -10 dB impedance bandwidths of 170 MHz and 370 MHz, mutual coupling below 20 dB, and fractional bandwidths of 12.2% and 14.6% at resonant frequencies of 1.3975 GHz and 2.45 GHz, respectively. Additionally, specific absorption rate (SAR) compliance and wireless telemetry performance assessments further validate the antenna suitability for biomedical applications.

I. INTRODUCTION

Recent advancements in medical electronic technologies have gained significant research interest in IMDs. These devices facilitate critical biomedical applications, including physiological monitoring, wireless data transmission, and neural stimulation, thereby enhancing early diagnosis and treatment. IMDs comprises several components, such as container, batteries, sensors, circuitry, and implantable antennas. Among these, the implantable antenna play significant role in establishing robust communication links between internal and external devices. However, the design of such antennas presents considerable challenges, including the imperative for miniaturization, multi-band functionality, wide bandwidth, gain, low SAR, biocompatibility, minimal coupling losses, reliable data transmission, and mitigation of frequency detuning induced by surrounding tissue heterogeneity. While conventional single-input-single-output (SISO) antenna structures [1–5] remain prevalent due to their simplicity and adequate performance, their limited data throughput renders them insufficient for next-generation IMDs, particularly those incorporating high-resolution imaging sensors that demand high-speed real-time video transmission. In contrast, MIMO antenna systems offer a transformative solution by enhancing channel capac-

ity, signal integrity, fading resilience, and overall communication reliability attributes essential for advanced real-time diagnostic applications. Nevertheless, key design challenges persist, including the suppression of inter-element coupling, impedance matching optimization, and further miniaturization to accommodate the compact size constraints of modern implantable devices. Recent studies have explored various MIMO antenna configurations to address the challenges of IMDs [6–9]. However, existing designs often suffer from critical limitations, including single-band operation, narrow bandwidth, and too large size, rendering them unsuitable for integration into modern compact IMDs. As the demand for minimally invasive implants continues to grow, there is an urgent need for miniaturized, multi-band MIMO antennas capable of operating within the constraints imposed by both biomedical implants and human anatomy. Such antennas must simultaneously achieve high performance while conforming to the dimensional and biocompatibility requirements of next-generation IMDs.

This study proposed a miniaturized dual-band implantable MIMO antenna designed specifically for leadless cardiac pacemaker system. The proposed antenna operates at two distinct frequencies 1.3975 GHz for wireless power transfer and 2.45 GHz for data telemetry, while maintaining compact form factor of 25.6 mm³. The four-element MIMO configuration demonstrates good inter-element isolation while meeting all critical performance characteristics, including reflection coefficient, gain, -10 dB impedance bandwidth, and fractional bandwidth. Comprehensive evaluations of SAR and wireless communication capabilities confirm the antenna usefulness for biomedical implantation. The design addresses the critical need for space-efficient, multi-functional antenna systems in next-generation implantable medical devices.

II. MIMO ANTENNA STRUCTURE AND DESIGN

Implantable antenna design presents unique challenges compared to conventional free-space antennas, requiring specialized simulation and measurement environments. Fig. 1 illustrates the 2D and 3D exploded views of both the SISO and MIMO antenna configurations. The proposed MIMO

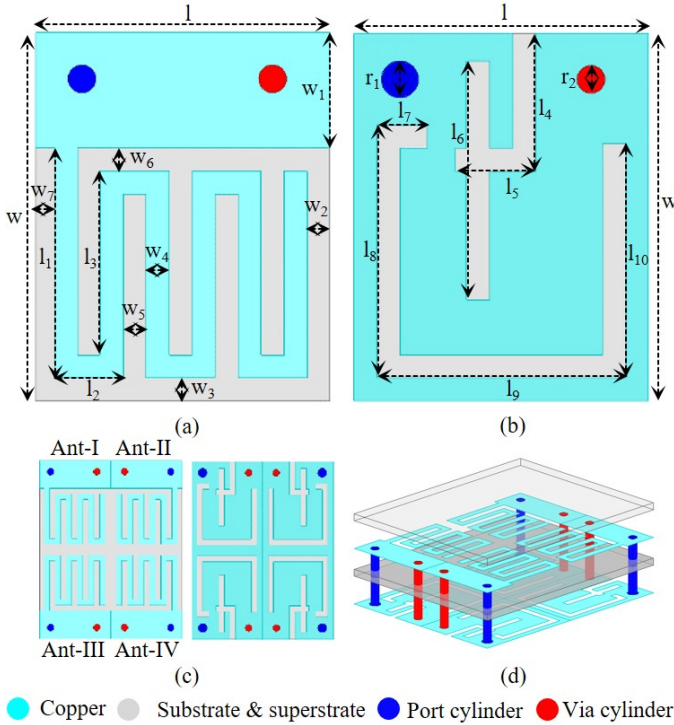


Fig. 1. (a) SISO patch (b) SISO ground (c) top and bottom view of MIMO (d) exploded 3D view of MIMO ($l = 4$, $w = 3.2$, $l_1 = 2.5$, $l_2 = 0.5$, $l_3 = 2.0$, $w_1 = 1.25$, $w_2 = w_3 = w_4 = w_5 = w_6 = 0.25$, $w_7 = 0.2$, $l_4 = 1.5$, $l_5 = 0.875$, $l_6 = 2.6$, $l_7 = 0.55$, $l_8 = 2.75$, $l_9 = 2.7$, $l_{10} = 2.55$, $r_1 = 0.2$, $r_2 = 0.15$).

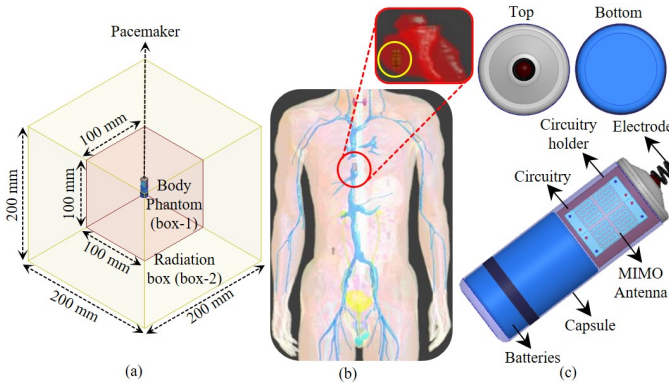


Fig. 2. (a) Homogeneous simulation setup (HFSS) (b) heterogeneous (Sim4life) (c) cardiac leadless pacemaker.

antenna consists of several key components: a shorting pin, a meandered radiating patch, a slotted ground plane, a substrate, and a superstrate. A high-dielectric constant material, Roger RT/duroid 6010, characterized by a dielectric permittivity of $\epsilon_r = 10.2$ and a loss tangent of $\tan\delta = 0.0023$, is employed for substrate and superstrate. Additionally, biocompatibility is achieved by encapsulating the antenna within a pacemaker capsule lined with alumina (Al_2O_3), which has a dielectric constant of $\epsilon_r = 9.8$ and a loss tangent of $\tan\delta = 0.006$, minimizing adverse effects on surrounding tissues. The pacemaker device has an diameter of 11 mm, a wall thickness of 0.25 mm,

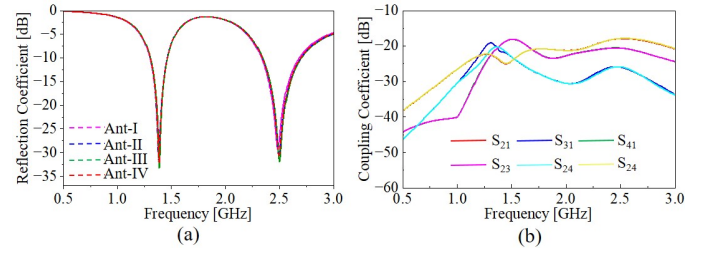


Fig. 3. (a) Reflection coefficient (b) coupling coefficient.

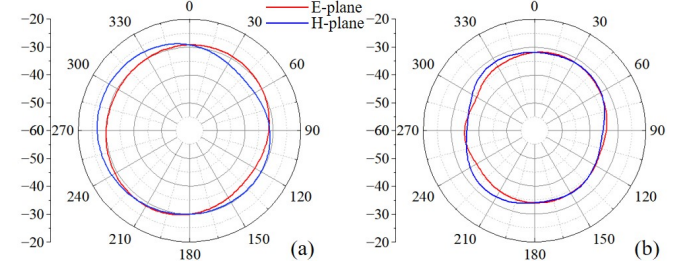


Fig. 4. Radiation pattern of Ant-I (a) 1.3975 GHz (b) 2.45 GHz.

and a length of 24.5 mm as shown in Fig. 2(c). A $50\ \Omega$ coaxial feed excites the antenna, which is initially simulated at the center of a homogeneous heart phantom (box-1: $100 \times 100 \times 100\ \text{mm}^3$) enclosed within a larger radiation boundary (box-2: $200 \times 200 \times 200\ \text{mm}^3$) as shown in Fig. 2(a). Given the spatial constraints of the human body, miniaturization is critical. The proposed MIMO antenna, with compact dimensions of $8 \times 6.4 \times 0.5\ \text{mm}^3$, employs meandering techniques to reduce size while maintaining performance. The proposed antenna system is implanted at a depth of 50 mm within the heart tissue phantom. To ensure biologically accurate simulation conditions, the dielectric properties of human cardiac tissue were applied to the phantom model (box-1) at both operational frequencies. The relative permittivity of the heart phantom measures 57.5 at 1.3975 GHz and 54.8 at 2.45 GHz, while the electrical conductivity exhibits values of 1.51 S/m and 2.26 S/m at these respective frequencies.

III. RESULTS AND DISCUSSION

The proposed MIMO antenna demonstrates stable frequency performance and low mutual coupling, making it suitable leadless cardiac pacemaker applications. As shown in Fig. 3 (a) all four antenna elements of MIMO antenna system consistently resonating within the intended frequency bands, with good reflection values below $-29\ \text{dB}$. All coupling coefficients are remain below 20 dB at both resonant frequencies, as illustrated in Fig. 3 (b). The antenna also achieved good bandwidths of 170 MHz and 370 MHz, with fractional bandwidths more than 12.4% and 14.6% at the lower and higher resonant frequencies, respectively. The simulated E-plane and H-plane of MIMO antenna radiation patterns at both frequencies, shown in Fig. 4. The gain is $-26.3\ \text{dBi}$ and $-30.15\ \text{dBi}$ at the lower and higher resonant frequencies, respectively. To validate the safety of the proposed MIMO antenna for biomedical applications, SAR

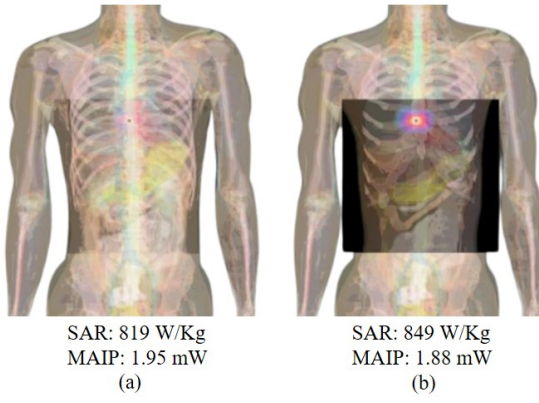


Fig. 5. Normalized 1-g SAR distribution and MAIP of the proposed MIMO antenna (a) 1.3975 GHz (b) 2.45 GHz.

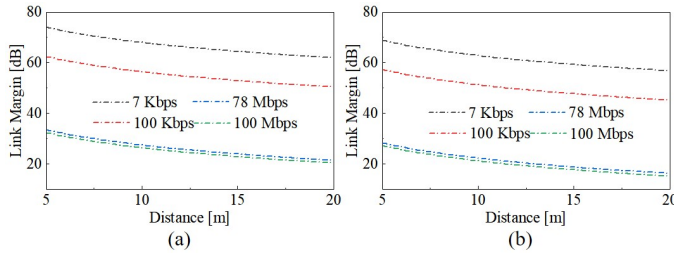


Fig. 6. Link margin analysis (a) 1.3975 GHz (b) 2.45 GHz.

values were evaluated at both operating frequencies. In compliance with the IEEE C95.1-2005 standard, the permissible SAR limits are 1.6 W/kg for 1-g of tissue and 2 W/kg for 10-g of tissue [10]. The antenna was simulated in a heterogeneous environment, retaining the same implant depth and positioning as in the homogeneous setup. The simulation, conducted using Sim4Life, is illustrated in Fig. 2 (b). Each antenna element was excited with 1 W of input power, and the resulting SAR values along with the maximum allowable input power (MAIP) are presented in Fig. 5. Given the input power limitation of -16 dBm for biomedical implants, the proposed antenna is confirmed to operate safely within the target frequency bands. The wireless telemetry capability of the proposed antenna was thoroughly evaluated through comprehensive transmission performance analysis. In the uplink communication scenario, the antenna operates as the transmitting T_x element while an external antenna serves as the receiving R_x component. As demonstrated in Fig. 6 (a) and (b), the MIMO antenna successfully maintains reliable high-speed communication up to 100 Mbps over several meters at both resonant frequencies, supporting various data rates including 7 Kbps, 100 Kbps, 78 Mbps, and 100 Mbps. Table 1 provides a comparison of the proposed MIMO antenna with recently reported implantable MIMO antennas for biomedical applications.

IV. CONCLUSION AND FUTURE WORK

A miniaturized dual-band implantable MIMO antenna has been developed for leadless cardiac pacemaker system, operating at 1.3975 GHz and 2.45 GHz. The compact design incor-

TABLE I
COMPARISON WITH RECENTLY PROPOSED IMPLANTABLE ANTENNAS

Ref.	Size [mm ³]	Freq [GHz]	Gain	BW [MHz]	Depth [mm]	Tissue
[6]	307	0.403	-36	145	4	1-Layer
[7]	441.27	2.45	-15.2	440	19.5	3-Layer
[8]	280.035	0.433	-21.8	210	3	1-Layer
[9]	3375	2.45, 5.8	-18.5	---	50	Muscle
This work	25.6	1.3975, 2.45	-26.3, -30.15	170, 370	50	Heart

porates four antenna elements with dimensions of $8 \times 6.4 \times 0.5$ mm³, yielding a total volume of just 25.6 mm³. Comprehensive simulations demonstrate good performance across key parameters including reflection coefficient, isolation, gain, bandwidth, and fractional bandwidth. SAR analysis confirms the antenna compliance with biomedical safety standards. Future work will focus on fabrication, experimental validation, and investigation of MIMO channel characteristics.

ACKNOWLEDGMENT

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